

Technology #09: UWB Catheter Position Mapping with SAR-Bounded Accuracy

Technology Domain: Radio Frequency (UWB) Localization + Tissue Propagation

Mechanism: Implantable UWB transmitter in shunt catheter; external receiver array localizes catheter position via time-of-arrival (TOA) ranging. R1 fix: original P-21 assumed sub-mm accuracy without SAR analysis; R1 redesign bounds accuracy by SAR limit and tissue propagation physics.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: SAR limit (1.6 W/kg) constrains transmit power below useful SNR at >5cm tissue depth

Next Experiment: UWB TX + RX link bench test

Buyer Receives: UWB localization concept + R1 correction (SAR-bounded accuracy); TOA + SAR + tissue propagation governing equations; Bench test protocols (link budget, SAR, accuracy, EMC, FCC); External precedent catalog (UWB medical radar, FCC Part 15.250, IEEE 1528); Critical UNKNOWN disclosure (achievable accuracy, antenna efficiency, multipath); This engineering dossier with failure analysis + build plan

Buyer Must Build: Production UWB transmitter design (IC selection, modulation, power management); Implantable UWB antenna design (small form factor); External receiver array (production hardware + localization software); Manufacturing process for catheter-integrated UWB implant; Regulatory submission (FDA + FCC combination; complex); Clinical validation (localization accuracy vs imaging); FCC certification + IEC 60601-1-2 compliance

Transfer Posture: Sponsored Validation

Target Buyer: Medical device localization company; neuroshunt manufacturer

CONFIDENTIAL. See full engineering dossier for details.